

Appendix B.1: Intermediate Lemmas

Lemma 1 (Gaussian ℓ_1 -norm tail bound). *Let $W \in \mathbb{R}^{m \times n}$ be a matrix with i.i.d. entries $W_{i,j} \sim \mathcal{N}(0, w^2)$. For any $\eta > 0$,*

$$\mathbb{P}(\|W\|_{\ell_1} \leq \eta) \geq 1 - \exp(-\eta/w + 2mn). \quad (1)$$

Proof. Let $Z = W_{i,j} \sim \mathcal{N}(0, w^2)$. Applying Markov's inequality to the exponentiated sum with $t = 1/w$:

$$\begin{aligned} \mathbb{P}(\|W\|_{\ell_1} > \eta) &= \mathbb{P}\left(\exp\left(\frac{1}{w}\|W\|_{\ell_1}\right) > \exp\left(\frac{\eta}{w}\right)\right) \\ &\leq \exp\left(-\frac{\eta}{w}\right) \prod_{i,j} \mathbb{E}[\exp(|Z|)]. \end{aligned} \quad (2)$$

Evaluating the moment-generating function of the standard half-normal distribution, $M_{|Z|}(t) = \exp(t^2/2)[1 + \operatorname{erf}(t/\sqrt{2})]$, at $t = 1$:

$$\mathbb{E}[\exp(|Z|)] = M_{|Z|}(1) = \exp(1/2) \left[1 + \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)\right] \approx 2.77 \leq e^2. \quad (3)$$

Substituting this loose upper bound yields:

$$\mathbb{P}(\|W\|_{\ell_1} > \eta) \leq \exp\left(-\frac{\eta}{w}\right) (e^2)^{mn} = \exp\left(-\frac{\eta}{w} + 2mn\right). \quad (4)$$

Taking the complement completes the proof. $\square$

Lemma 2 (Lipschitz smooth bound for FTD). *Let the tensor $\mathcal{X}_t \in \mathbb{R}^{I_1^t \times I_2^t \times I_3^t}$ satisfy FTD, where each factor function is of the form $f_{\Theta_n}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^r$ ($n = 1, 3$) and $f_{\Theta_2}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^2$ where both are INR's formulated as $f_{\Theta_n}(v) = W_d(\sigma(W_{d-1} \cdots \sigma(W_1 v)))$ with activation function $\sigma(\cdot) = \sin(\omega_0 \cdot)$. Assume that each element of the weight matrix W follows i.i.d. $\mathcal{N}(0, w^2)$. Then for any $\delta \in (0, 1)$, any spatial coordinates $v, v' \in \mathbb{Z}^2$, where $v = (v_{(1)}, v_{(2)})$, $v' = (v'_{(1)}, v'_{(2)})$ and any temporal coordinates $k, k' \in \mathbb{Z}$, with probability at least $1 - \delta$ the following Lipschitz smoothness holds for $\mathcal{X}_t$:*

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v'_{(1)}, v'_{(2)}, k)}| \leq C_1 \|v - v'\|_{l_1}, \quad (\text{Spatial smooth}) \quad (5)$$

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k')}| \leq C_1 |k - k'|, \quad (\text{Temporal smooth}) \quad (6)$$

where $C_1 = \omega_0^{3d-3} (2wr^3 + w \ln \frac{3d}{\delta})^{3d} \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$ is a Lipschitz constant.

Proof. Since each element of the weight matrices $W \in \mathbb{R}^{r \times r}$ follows i.i.d. $\mathcal{N}(0, w^2)$, we have the tail bound

$$\mathbb{P}(\|W\|_{l_1} \leq \eta) \geq 1 - \exp(-\eta/w + 2r^2) \quad (7)$$

We denote d to be the number of layers in and INR so we have $2d$ layers from $f_{\Theta_1}, f_{\Theta_3}$ with each layer $W \in \mathbb{R}^{r \times r}$ and another $d-1$ layers with $W \in \mathbb{R}^{r \times r}$ and a single layer with $W \in \mathbb{R}^{r \times r^2}$. Hence from the previous lemma the probability of all the weight matrices of the OFTD model have l_1 -norm less than η is at least

$$1 - (3d-1) \exp(-\eta/w + 2r^2) - \exp(-\eta/w + 2r^3) \geq 1 - 3d \exp(-\eta/w + 2r^3)$$

Let $1 - 3d \exp(-\eta/w + 2r^3) = 1 - \delta$, we have $\eta = 2wr^3 + w \ln \frac{3d}{\delta}$.

Note that the sine activation function satisfies $|\sin(\omega_0 x)| \leq \omega_0 |x|$ ($\omega_0 > 0$). Hence we have $\|f_{\Theta_n}(x)\|_{l_1} \leq \eta^d \omega_0^{d-1} |x|$. For any (i, j, k) we have

$$(\mathcal{X}_t)_{(i,j,k)} = f_{\Theta_1}([I_1^t]_{(i)}^T) f_{\Theta_2}([I_2^t]_{(j)}) f_{\Theta_3}([I_3^t]_{(k)}). \quad (8)$$

Where $f_{\Theta_1}([I_1^t]_{(i)}), f_{\Theta_3}([I_3^t]_{(k)}) \in \mathbb{R}^r$ and $f_{\Theta_2}([I_2^t]_{(j)}) \in \mathbb{R}^{r \times r}$. We first note that $f_{\Theta_2}([I_2^t]_{(j)}) \in \mathbb{R}^{r \times r}$ is a matrix and we can get the following loose upper bound:

$$\|f_{\Theta_2}([I_2^t]_{(j)})\|_{op} \leq \|f_{\Theta_2}([I_2^t]_{(j)})\|_{l_1}$$

We will use this loose upper bound in our proof to match our union bound we derived. Then for any (i, j, k) and (i', j, k) we have

$$\begin{aligned} |(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i',j,k)}| &= \left| (f_{\Theta_1}([I_1^t]_{(i)}^T) - f_{\Theta_1}([I_1^t]_{(i')}^T)) f_{\Theta_2}([I_2^t]_{(j)}) f_{\Theta_3}([I_3^t]_{(k)}) \right| \\ &\leq \|f_{\Theta_1}([I_1^t]_{(i)}^T) - f_{\Theta_1}([I_1^t]_{(i')}^T)\|_{l_1} \|f_{\Theta_2}([I_2^t]_{(j)})\|_{l_1} \|f_{\Theta_3}([I_3^t]_{(k)})\|_{l_1} \\ &= \|f_{\Theta_1}(i) - f_{\Theta_1}(i')\|_{l_1} \|f_{\Theta_2}(j)\|_{l_1} \|f_{\Theta_3}(k)\|_{l_1} \end{aligned} \quad (9)$$

The bound on the first term holds as:

$$\begin{aligned} \|f_{\Theta_1}(i) - f_{\Theta_1}(i')\|_{l_1} &= \|W_d(\sigma(W_{d-1} \cdots \sigma(W_1 i))) - W_d(\sigma(W_{d-1} \cdots \sigma(W_1 i')))\|_{l_1} \\ &\leq \eta \|\sigma(W_{d-1} \cdots \sigma(W_1 i)) - \sigma(W_{d-1} \cdots \sigma(W_1 i'))\|_{l_1} \\ &\leq \eta \omega_0 \|W_{d-1} \cdots \sigma(W_1 i) - W_{d-1} \cdots \sigma(W_1 i')\|_{l_1} \\ &\leq \eta^d \omega_0^{d-1} |i - i'|. \end{aligned} \quad (10)$$

Thus we have

$$\begin{aligned} |(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i',j,k)}| &\leq \|f_{\Theta_1}(i) - f_{\Theta_1}(i')\|_{l_1} \|f_{\Theta_2}(j)\|_{l_1} \|f_{\Theta_3}(k)\|_{l_1} \\ &\leq \eta^d \omega_0^{d-1} |i - i'| \eta^d \omega_0^{d-1} |j| \eta^d \omega_0^{d-1} |k| \\ &\leq \eta^{3d} \omega_0^{3d-3} I_2^t I_3^t |i - i'|. \end{aligned} \quad (11)$$

Similarly, we can derive the analogous bounds for the two other dimensions:

$$|(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i,j',k)}| \leq \eta^{3d} \omega_0^{3d-3} I_1^t I_3^t |j - j'| \quad (12)$$

$$|(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i,j,k')}| \leq \eta^{3d} \omega_0^{3d-3} I_1^t I_2^t |k - k'| \quad (13)$$

Hence from the triangle inequality one can easily derive that for any spatial coordinates v, v' and temporal coordinates k, k' the following bounds hold with probability at least $1 - \delta$:

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v'_{(1)}, v'_{(2)}, k)}| \leq C_1 \|v - v'\|_{l_1}, \quad (14)$$

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k')}| \leq C_1 |k - k'|, \quad (15)$$

where $C_1 = \omega_0^{3d-3}(2wr^3 + w \ln \frac{3d}{\delta})^{3d} \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$. The proof is completed. $\square$

Appendix B.2: Proof of Online FTD

Theorem 1 (Regret bound for online FTD). *Denote the OFTD model learned at the historical time point t by $\mathcal{X}_t = [[\{f_{\Theta_n}([I_n^t]; t)\}_{n=1}^3]] \in \mathbb{R}^{I_1^t \times I_2^t \times I_3^t}$, where $f_{\Theta_n}([I_n^t]; t)$ denotes the n -th factor function at time t . Assume that:*

- *The OFTD model learned at the new time point $t + 1$ is invariant at the boundary (I_1^t, I_2^t, I_3^t) , i.e., $f_{\Theta_n}(I_n^t; t + 1) = f_{\Theta_n}(I_n^t; t)$ ($n = 1, 2, 3$).*
- *Each element of the weight matrix of the d -layer INRs $\{f_{\Theta_n}(\cdot)\}$ follows i.i.d. $\mathcal{N}(0, w^2)$ with sine activation function $\sigma(\cdot) = \sin(\omega_0 \cdot)$.*
- *The l_1 -norm of derivative of each factor INR $\|f'_{\Theta_n}(x)\|_{l_1}$ is bounded by $\kappa > 0$.*

Then for any $\delta \in (0, 1)$ and any historical position (i_1, i_2, i_3) ($i_n \leq I_n^t$), the following regret bound between the new OFTD model $\mathcal{X}_{t+1} = [[\{f_{\Theta_n}([I_n^{t+1}]; t + 1)\}_{n=1}^3]] \in \mathbb{R}^{I_1^{t+1} \times I_2^{t+1} \times I_3^{t+1}}$ (Take an example $I_n^{t+1} = I_n^t + 1$) and the old OFTD model $\mathcal{X}_t$ holds with probability at least $1 - \delta$:

$$|(\mathcal{X}_{t+1})_{(i_1, i_2, i_3)} - (\mathcal{X}_t)_{(i_1, i_2, i_3)}| \leq C_2 \max_n (I_n^t + 1 - i_n), \quad (16)$$

where $C_2 = 6(\eta^{3d} \omega_0^{3d-3} + \kappa \eta^{2d} \omega_0^{2d-2}) \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$ and $\eta = 2wr^3 + w \ln \frac{3d}{\delta}$.

Proof. First we derive the regret bound for each factor INR $f_{\Theta_n}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^r$. For any $x \in [0, I_n^t]$ we have:

$$\begin{aligned} \|f_{\Theta_n}(x; t + 1) - f_{\Theta_n}(x; t)\|_{l_1} &\leq \left\| f_{\Theta_n}(x; t + 1) - \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t + 1) ds \right\|_{l_1} \\ &\quad + \left\| \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t + 1) ds - \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t) ds \right\|_{l_1} \\ &\quad + \left\| \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t) ds - f_{\Theta_n}(x; t) \right\|_{l_1}. \end{aligned} \quad (17)$$

From Lemma 1 and its proof, we have that with probability at least $1 - \delta$ the first term on the right-hand side of (17) admits:

$$\begin{aligned}
\left\| f_{\Theta_n}(x; t+1) - \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t+1) ds \right\|_{l_1} &= \left\| \int_{I_n^t}^{I_n^{t+1}} (f_{\Theta_n}(x; t+1) - f_{\Theta_n}(s; t+1)) ds \right\|_{l_1} \\
&\leq \left\| \int_{I_n^t}^{I_n^{t+1}} \eta^d \omega_0^{d-1} (s - x) ds \right\|_{l_1} \\
&\leq \left\| \int_{I_n^t}^{I_n^{t+1}} \eta^d \omega_0^{d-1} (I_n^t + 1 - x) ds \right\|_{l_1} \\
&= \eta^d \omega_0^{d-1} |I_n^t + 1 - x|, \tag{18}
\end{aligned}$$

where $\eta = 2wr^2 + w \ln \frac{3d}{\delta}$. Similarly we can get for the third term:

$$\left\| \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t) ds - f_{\Theta_n}(x; t) \right\|_{l_1} \leq \eta^d \omega_0^{d-1} |I_n^t + 1 - x| \tag{19}$$

Define $h_n(x) = f_{\Theta_n}(x; t+1) - f_{\Theta_n}(x; t)$, and from assumptions we have $h_n(I_n^t) = 0$. From the mean value theorem for integrals, the second term on the right-hand side of (17) admits:

$$\left\| \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t+1) ds - \int_{I_n^t}^{I_n^{t+1}} f_{\Theta_n}(s; t) ds \right\|_{l_1} = \left\| \int_{I_n^t}^{I_n^{t+1}} h_n(s) ds \right\|_{l_1} = \|h_n(\zeta)\|_{l_1} \tag{20}$$

where $\zeta \in (I_n^t, I_n^t + 1)$. By the Lagrange mean value theorem we have:

$$\|h_n(\zeta)\|_{l_1} = \|h_n(\zeta) - h_n(I_n^t)\|_{l_1} = \|h'_n(\xi)(\zeta - I_n^t)\|_{l_1} \leq 2\kappa |I_n^t + 1 - x| \tag{21}$$

where $\xi \in (I_n^t, \zeta)$. Here, we have used the assumption that $\|h'_n(\xi)\|_{l_1} = \|f'_{\Theta_n}(\xi; t+1) - f'_{\Theta_n}(\xi; t)\|_{l_1} \leq 2\kappa$. Combining all the results, we obtain the regret bound for the individual factor INRs:

$$\|f_{\Theta_n}(x; t+1) - f_{\Theta_n}(x; t)\|_{l_1} \leq 2(\eta^d \omega_0^{d-1} + \kappa) |I_n^t + 1 - x|. \tag{22}$$

Now we can derive the regret bound for online FTD. Write each element of the tensor $\mathcal{X}_t$ as

$$(\mathcal{X}_t)_{(i_1, i_2, i_3)} = f_{\Theta_1}([I_1^t]; t)_{(i_1)} f_{\Theta_2}([I_2^t]; t)_{(i_2)} f_{\Theta_3}([I_3^t]; t)_{(i_3)}$$

By adding and subtracting terms (standard product difference expansion), for

any (i_1, i_2, i_3) ($i_n \leq I_n^t$) we have:

$$\begin{aligned}
& |(\mathcal{X}_{t+1})_{(i_1, i_2, i_3)} - (\mathcal{X}_t)_{(i_1, i_2, i_3)}| \\
&= |f_{\Theta_1}([I_1^t]; t+1)_{(i_1)}^T f_{\Theta_2}([I_2^t]; t+1)_{(i_2)} f_{\Theta_3}([I_3^t]; t+1)_{(i_3)} - f_{\Theta_1}([I_1^t]; t)_{(i_1)}^T f_{\Theta_2}([I_2^t]; t)_{(i_2)} f_{\Theta_3}([I_3^t]; t)_{(i_3)}| \\
&\quad (23) \\
&= |(f_{\Theta_1}(i_1; t+1) - f_{\Theta_1}(i_1; t))^T f_{\Theta_2}(i_2; t) f_{\Theta_3}(i_3; t) \\
&\quad + f_{\Theta_1}(i_1; t+1)^T (f_{\Theta_2}(i_2; t+1) - f_{\Theta_2}(i_2; t)) f_{\Theta_3}(i_3; t) \\
&\quad + f_{\Theta_1}(i_1; t+1)^T f_{\Theta_2}(i_2; t+1) (f_{\Theta_3}(i_3; t+1) - f_{\Theta_3}(i_3; t))| \\
&\leq \|f_{\Theta_1}(i_1; t+1) - f_{\Theta_1}(i_1; t)\|_{l_1} \|f_{\Theta_2}(i_2; t)\|_{l_1} \|f_{\Theta_3}(i_3; t)\|_{l_1} \\
&\quad + \|f_{\Theta_2}(i_2; t+1) - f_{\Theta_2}(i_2; t)\|_{l_1} \|f_{\Theta_1}(i_1; t+1)\|_{l_1} \|f_{\Theta_3}(i_3; t)\|_{l_1} \\
&\quad + \|f_{\Theta_3}(i_3; t+1) - f_{\Theta_3}(i_3; t)\|_{l_1} \|f_{\Theta_1}(i_1; t+1)\|_{l_1} \|f_{\Theta_2}(i_2; t+1)\|_{l_1}. \quad (24)
\end{aligned}$$

Substituting (22) and the bound we derived in the previous lemma $\|f_{\Theta_n}(x)\|_{l_1} \leq \eta^d \omega_0^{d-1} |x|$ into (24) we obtain:

$$\begin{aligned}
& |(\mathcal{X}_{t+1})_{(i_1, i_2, i_3)} - (\mathcal{X}_t)_{(i_1, i_2, i_3)}| \\
&\leq 2(\eta^d \omega_0^{d-1} + \kappa) |I_1^t + 1 - i_1| (\eta^d \omega_0^{d-1})^2 I_2^t I_3^t \\
&\quad + 2(\eta^d \omega_0^{d-1} + \kappa) |I_2^t + 1 - i_2| (\eta^d \omega_0^{d-1})^2 I_1^t I_3^t \\
&\quad + 2(\eta^d \omega_0^{d-1} + \kappa) |I_3^t + 1 - i_3| (\eta^d \omega_0^{d-1})^2 I_1^t I_2^t \\
&\leq C_2 \max_n (I_n^t + 1 - i_n) \quad (25)
\end{aligned}$$

where $C_2 = 6(\eta^{3d} \omega_0^{3d-3} + \kappa \eta^{2d} \omega_0^{2d-2}) \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$. The proof is completed. $\square$